

Prerequisites for AI

Basic Computer Concepts

Main Components of a Computer :-Hardware,Software,Input Devices,Output Devices,Storage Devices
Internet & Browser Usage Common Internet Uses Popular Web Browsers

Searching Information Using a Browser

Email & Online Accounts

Functions of Email

Common Online Accounts

Practical Example: Sending an Email

Cloud Storage Basics

Advantages of Cloud Storage

Examples of Cloud Storage Services

Digital Safety & Password Management

Understanding Data & Logic

What is Data?

Importance of Data

Practical Example: Student Attendance Data

Types of Data (Text, Image, Audio, Video)

Practical Example: Data Types in a Smartphone

Data Collection Methods

Practical Example: Collecting Online Registration Data

4. Data Accuracy & Quality

Factors Affecting Data Quality Accuracy ,Completeness ,Consistency

Timeliness ,Relevance

Basics of Logic & Decision Making

Introduction to Algorithms (Simple Explanation)

- Introduction to AI
- what is AI?
- History & Evolution of AI
- AI vs Human Intelligence
- Real-World Examples of AI
- Why is AI Important?
- How does AI work?
- Types of AI
- **Future of AI**
- **APPLICATIONS OF AI**
- **KEY COMPONENTS OF AI**

what is AI?

Key Characteristics of Artificial Intelligence

- Ability to learn from data
- Ability to reason and make decisions
- Ability to solve problems
- Ability to adapt to new situations

Practical Example: Simple AI-Based Decision Making

Example: AI deciding whether an email is spam or not.

Steps:

- The system collects thousands of emails as data.
- It studies patterns such as words, links, and sender details.
- The AI learns which patterns usually appear in spam emails.
- When a new email arrives, the AI compares it with learned patterns.
- The system classifies the email as spam or normal.

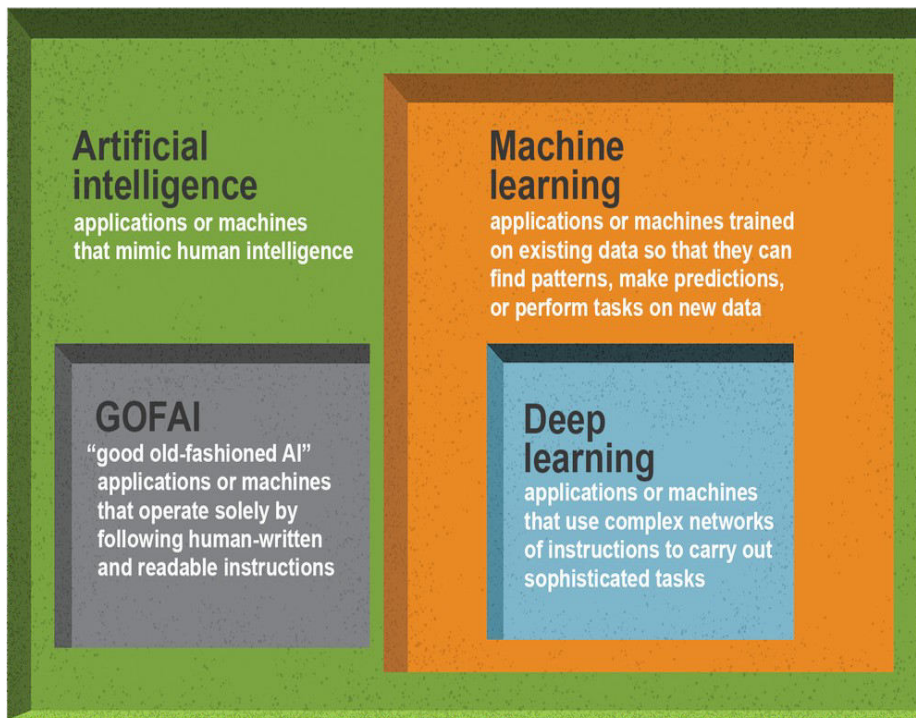
History & Evolution of AI

Major Phases in AI Evolution

- **1950s:** Concept of machine intelligence introduced.
- **1960s–1970s:** Early rule-based AI systems developed.
- **1980s:** Expert systems used in industries.
- **1990s–2000s:** Machine learning and data-based systems improved.
- **2010 onwards:** Deep learning, automation, and smart applications expanded rapidly.

Difference Between AI, ML & Deep Learning

Artificial Intelligence, Machine Learning, and Deep Learning are related but not identical concepts.



Artificial Intelligence

Broad field of creating intelligent systems
Includes rule-based and learning systems

Machine Learning

Subset of artificial intelligence

Focuses on learning from data

Deep Learning

Subset of machine learning

Uses layered structures to process data

1 Practical Example: Image Recognition Comparison

Steps:

- AI system identifies an image using rules.
- Machine learning system learns features from data.
- Deep learning system processes image layers.
- Accuracy improves with deeper analysis.
- Each level offers increasing complexity and performance.

2 Practical Example: Evolution of Search Engines

2000 -- 2010 -- 2020 --2030

Steps:

- Early search engines showed results based on exact keyword matching.
- Later systems analyzed page relevance and popularity.
- Modern AI-based search engines understand user intent.
- They learn from user behavior and improve results continuously.
- Results become more accurate and personalized over time.

Real-World Examples of AI

1. Virtual Assistants (e.g., Siri, Alexa)
2. Self-Driving Cars (e.g., Tesla)
3. Recommendation Systems (e.g., Netflix, Amazon)
4. Facial Recognition Software
5. Maps and Navigation
6. Text Editors and Autocorrect (e.g., Grammarly)
7. Chatbots: AI-Powered Customer Support
8. E-Payments: AI in Banking
 - Convenience
 - Fraud Detection

Practical Example: AI in Online Shopping

Steps:

- A user searches for a product online.
- The system records browsing and purchase history.
- AI analyzes user preferences and behavior.
- Similar product suggestions are shown.
- Recommendations improve as user activity increases.

Why is AI Important? Need of AI?

1. Simplifies Daily Tasks
2. Solves Complex Problems
3. Increases Efficiency
4. Enhances Personalization
5. Boosts Business Efficiency
6. Improves Healthcare
7. Enables Smarter Transportation
8. Facilitates Better Decision-Making
9. Strengthens Security
10. Supports Education
11. Advances Scientific Research
12. Improves Accessibility
13. Drives Innovation

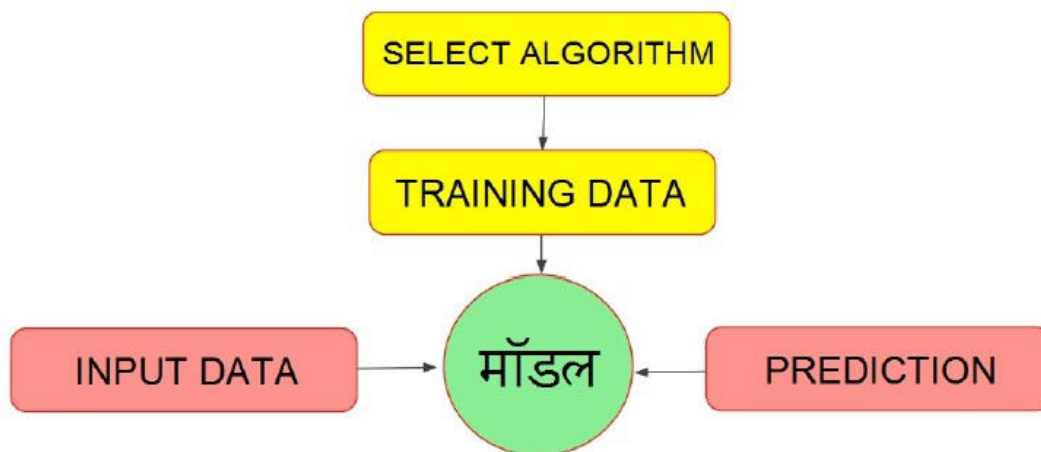
How does AI work?

Understanding Machine Learning

Key Features of Machine Learning

- Learning from data
- Pattern recognition
- Performance improvement over time
- Data-driven decision making

Key Concepts of Machine Learning



Key Concepts of Machine Learning

1. Training Data

Example:

2. Algorithms

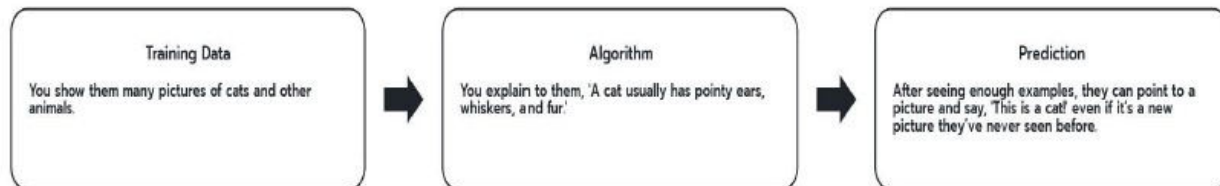
Example:

3. Predictions

Example:

A Simple Analogy: Teaching a Child to Recognize a Cat

Imagine you want to teach a child how to recognize a cat. Here's how it would work:



Types of Machine Learning

1. Supervised ML (SUITABLE FOR PREDICTION AND CLASSIFICATION)

Supervised learning is a type of machine learning where the system is trained using **labeled data**.

Each input is provided with the correct output, allowing the system to learn by example

Practical Example: Student Result Prediction

2. Unsupervised ML (SUITABLE FOR DATA GROUPING AND CLUSTERING)

Unsupervised learning uses **unlabeled data**.

The system identifies hidden patterns and groupings without predefined outputs.

Practical Example: Customer Segmentation

3. Reinforcement ML (SUITABLE FOR CONTINUOUS IMPROVEMENT)

Reinforcement learning is a type of machine learning where a system learns through **trial and error**.

The system receives feedback in the form of rewards or penalties based on actions taken.

Practical Example: Game Playing System

Real-Life ML Examples

Machine learning is widely used in everyday applications to improve accuracy, efficiency, and user experience.

Common Real-Life Applications

- Email filtering
- Voice recognition
- Recommendation systems
- Fraud detection
- Traffic prediction

Practical Example: Voice Recognition System

Steps:

1. User speaks a command.
2. Audio data is captured.
3. The system analyzes sound patterns.
4. Speech is converted into text.
5. The system responds accordingly.

AI vs Human Intelligence

Natural and emotional

HI :-Human Intelligence

Human intelligence is creative and emotional.

Humans can think independently.

Strengths of Human Intelligence

- Emotional understanding
- Moral judgment
- Creativity and imagination

artificial and logical

AI :-Artificial Intelligence

AI intelligence is data-driven and logical.

AI works only within its programmed and learned limits.

Strengths of AI

- Works continuously without fatigue
- Handles large data accurately
- Performs repetitive tasks efficiently

Practical Example: AI Calculator vs Human Calculation

Steps:

- A human solves a complex calculation manually.
- Time and chances of error increase with complexity.
- An AI calculator processes the same calculation instantly.
- AI gives accurate results every time.
- Human intelligence is still required to decide when and how to use results.

Types of AI

Introduction to Different Types of AI

1. Narrow AI
2. General AI
3. Super AI

1. Narrow AI (Weak AI)

Definition

Narrow AI refers to AI systems that are designed to handle a specific task or solve a particular problem.

It's called "weak" AI

Key Characteristics:

Task-specific

No self-awareness

Examples(VOICE ASSISTANTS AND RECOMMENDATION)

Siri or Alexa

Recommendation Systems (Netflix, YouTube)

Self-driving Cars

2. General AI (Strong AI)

Definition:

General AI, also known as strong AI

Key Characteristics:

Versatility

Self-awareness

Examples (theoretical)

A human-like robot

Note: General AI is still largely a concept and has not yet been achieved. Researchers are still working toward creating AI that can perform a wide range of tasks and think more like humans.

3)Super AI or Artificial Superintelligence (ASI)

Definition

is a theoretical tier of machine intelligence that completely surpasses human capabilities across all intellectual domains.

Key Characteristics:

Self-Improvement Loops

Multi-Domain Mastery

Advanced Problem Solving

Superhuman Creativity

Social and Emotional Manipulation

Examples (theoretical)

Because ASI does not yet exist, its examples are found in scientific concepts and philosophical thought experiments

- *The Technological Singularity*
- *The Cosmic Explorer*
- *Global Optimization Network*
- *Sci-Fi Archetypes*

Practical Example: Identifying AI Types

Steps:

- A voice assistant answers questions and sets reminders.
- It performs only limited predefined tasks.
- This system is classified as Narrow AI.
- A hypothetical system that thinks and learns like a human would be General AI.
- A system that improves beyond human intelligence would be Super AI.

FUTURE OF AI

1. AI in Healthcare

Personalized Medicine
Medical Imaging and Diagnostics
Virtual Health Assistants

2. AI in Education

- *Personalized Learning*
- *Smart Classrooms*
- *Global Learning Access*
- *Content Creation and Curation*
- *Language and Skill Learning*

3. AI for the Environment

- *Climate Change Monitoring*
- *Sustainable Agriculture*
- *Wildlife Protection*
- *Energy Efficiency*

4. AI in Transportation

- Self-Driving Cars
- Smart Traffic Management
- Automated Delivery Systems
- Predictive Maintenance
- Fleet Management
- Ride-Sharing Optimization
- Air Traffic Control

5. AI in Business and Economy

- Automation of Routine Tasks
- Data-Driven Decision Making
- Supply Chain Optimization

6. AI for Creativity and Entertainment

- AI in Art and Design:
- Personalized Entertainment Recommendations:
- Virtual Content Creation:
- Interactive Storytelling:
- AI-Generated Writing:
- Real-Time Translation and Subtitling:
- Music Composition and Mixing:
- Immersive Experiences with AR/VR

7. AI in Security and Law Enforcement

- Predictive Policing:
- Cybersecurity:
- Surveillance:
- Fraud Detection:
- Missing Person Identification:
- Traffic Law Enforcement:
- Threat Assessment:
- Emergency Response:
- Behavioral Analysis:

Practical Example: AI in Future Education Systems Steps:

- Student learning data is collected digitally.
- AI analyzes strengths and weaknesses.
- Personalized learning content is generated.
- Performance is monitored continuously.
- Learning outcomes improve through adaptive methods.

- **APPLICATIONS OF AI**

- Gaming
- Natural Language Processing
- Expert Systems
- Vision Systems
- Speech Recognition

- Handwriting Recognition
- Intelligent Robots

KEY COMPONENTS OF AI

1. Technological Engines (How AI Learns)

- Machine Learning (ML)
- Deep Learning
- Algorithms

2. Data & Infrastructure (The Fuel and Muscle)

- Data
- Hardware & Cloud Infrastructure

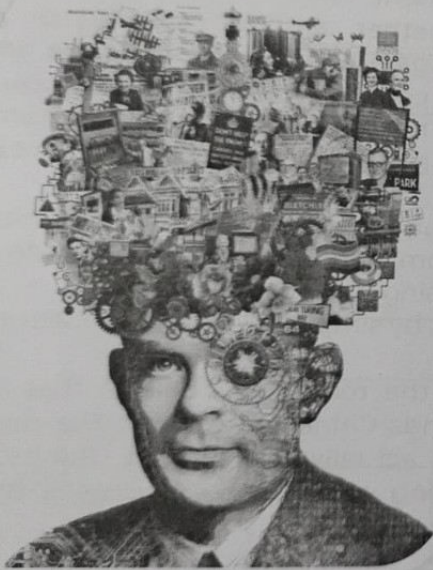
3. Cognitive Capabilities (How AI Interacts)

- Natural Language Processing (NLP)
- Computer Vision
- Automated Reasoning

4. Physical & Operational Layers (How AI Deploys)

- Robotics
- Ethics & Governance

Acting Like a Human: The Turing Test



The **Turing Test** is a test proposed in 1950 by British mathematician and computer scientist **Alan Turing**, designed to evaluate whether a machine can exhibit intelligent behavior equivalent to, or indistinguishable from, that of a human.

In this test:

- A **judge** (a human) communicates in writing with both a human and a machine (typically a computer program),
- The judge tries to determine which is the human and which is the machine,
- If the judge cannot reliably distinguish the machine's responses from the human's, the machine is said to have passed the Turing Test.

What Can't Artificial Intelligence Do?

The "argument from disability" makes the claim that "a machine can never do X."

As examples of X, Turing lists the following:

- Be kind,
- resourceful,
- beautiful,
- friendly,
- have initiative,
- have a sense of humor,
- tell right from wrong,
- make mistakes,
- fall in love,
- enjoy strawberries and cream,
- make someone fall in love with it,
- learn from experience,
- use words properly,
- be the subject of its own thought,
- have as much diversity of behavior as man,
- do something really new.

Assignment 1: use ASP.net core or form-based (create a panel for products - admin end)
-Asp.net core with mvc template would be preferred -

use database

The form is titled "Submit a Product" and contains the following fields and controls:

- Product Name:** A text input field.
- Product Category:** A dropdown menu with "Please Select" as the current selection.
- Product Freshness:** Three radio buttons labeled "Brand New", "Second Hand", and "Refurbished".
- Image of Product:** A "Choose File" button and a text label "No file chosen".
- Additional Description:** A large text area.
- Product Price:** A text input field with "ex: \$5" as a placeholder, followed by an "MRP" label and another text input field.
- Comments:** A large text area.
- Manager email:** A text input field.
- Manager Phone number:** A text input field.
- Submit:** A button at the bottom center.

should have grid to show data with create insert update and delete view do crud operation using entity framework

- create crud operation for this
- Create the above form and insert data into SQL Server database.
- Add mandatory field validation to all fields. Product price and mrp fields should have numeric value validation. Email and phone number should have respective validations.
- Show the inserted data in a table. Show All fields (including image uploaded from the form).
- Provide a button to delete each entry in the table and on click delete the entry.

Assignment 2 :-

Task Description for .NET Developer

Objective: Create a Survey Page with conditional (if/else) flow of questions based on user responses.

Details:

- Build a simple survey web page using .NET (ASP.NET MVC Web Forms).
- Survey will have 10 main questions.
- Based on the user's answer (Yes/No), the next question will change dynamically.
- After each branch, the survey should continue normally.
- Use a circuit/flow structure to manage the conditional logic between questions.
- All questions and options must be fetched dynamically from the database.

Expectations:

- Proper structure for navigating between questions based on responses.
- Clean UI (no need for heavy CSS, just basic neat layout).
- Questions and options should be stored in the database and fetched dynamically.
- Proper validation (like required answers where necessary).
- Maintain session state for responses.
- Try submit all answers at the end.

[Q1: Are you married?]

└─ Yes → [Q2: How many children do you have?]

| ↓

| [Q3: How was your day?]

└─ No → [Q3: How was your day?]

[Q4: Do you own a car?]

└─ Yes → [Q5: What is the make and model of your car?]

└─ No → [Q5: Do you plan to buy one in the next year?]

[Q6: Are you currently employed?]

└─ Yes → [Q7: What is your job title?]

└─ No → [Q7: Are you currently looking for a job?]

[Q8: Do you work remotely?]

└─ Yes → [Q9: What tools do you use for remote work?]

└─ No → [Q9: How long is your daily commute?]

[Q10: Do you exercise regularly?]

└─ Yes → [Q11: What type of exercise do you prefer?]

└─ No → [Q11: What's stopping you from exercising?]